CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
First Semester of M.Sc. (Mathematics) Examination March 2018
MA711 Complex Analysis

Date: 26/03/2018, Monday Time: 10:00 a.m. to 10:45 a.m. Maximum Marks: 20

MCQ

Important Instructions:

1. Tick the correct answer and it should be written in question paper itself.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Each question carries one mark.
4. All notations are standard.

Q - I Choose the correct answer from the given options for the following questions.

1. Log(-1)= ______.

(a) πi **(b)** ∞ **(c)** $2\pi i$ **(d)** 0

2. The function $f(z) = u(x, y) + iv(x, y)$ is differentiable at $z_0 = x_0 + iy_0$ then $f'(z_0) =$ ______.

(a) $u_x(x_0, y_0) + v_x(x_0, y_0)$ **(c)** $u_y(x_0, y_0) + v_x(x_0, y_0)$
(b) $u_x(x_0, y_0) - v_x(x_0, y_0)$ **(d)** None of these

3. $\mathrm{Arg}(-1 - i) =$ ______.

(a) $\frac{1}{4}\pi$ **(b)** $-\frac{1}{4}\pi$ **(c)** $-\frac{5}{4}\pi$ **(d)** $-\frac{3}{4}\pi$

4. The function $u(x,y)$ is harmonic if ______.

(a) $u_x + u_y = 0$ (b) $u_{xy} + u_{xy} = 0$ (c) None of these (d) $u_{xx} + u_{yy} = 0$

5. The radius of convergence of $\sum_{n=0}^{\infty}(4n+1)z^n$ is ______.

(a) 4 (b) ∞ (c) 0 (d) 1

6. For any $z \in \mathbb{C}$, $|e^z| =$ ______.

(a) e^x (b) 1 (c) e^y (d) $2\pi i$

7. If $z_1, z_2 \in \mathbb{C}$ with $z_2 \neq 0$ then $\log\left(\frac{z_1}{z_2}\right) =$ ______.

(a) $\log z_1 + \log z_2$ (b) $\log z_1 \log z_2$ (c) $\log z_1 - \log z_2$ (d) $\log z_2 / \log z_1$

8. The isolated singular point(s) of the function $\dfrac{1}{\sin(\pi/z)}$ is(are) ______.

(a) $n\pi$, $n \in \mathbb{N}$

(b) $\frac{1}{n}$, $n \in \mathbb{N}$

(c) 0

(d) $(2n+1)\frac{\pi}{2}$, $n \in \mathbb{N}$

9. The Cauchy-Riemann equations are given by ______.

(a) $u_x = v_y, u_y = v_x$

(b) $u_x = v_x, u_y = v_x$

(c) $u_x = v_y, u_y = v_y$

(d) $u_x = v_y, u_y = -v_x$

10. If C denotes the positively oriented circle $|z - z_0| = R$, then the value of $\int_C (z - z_0)^2 \mathrm{d}z$ is ______.

(a) 0 (b) $2\pi i$ (c) 2π (d) $\frac{1}{2\pi i}$

11. A non-constant entire function is ______.

(a) discontinuous

(b) bounded

(c) unbounded

(d) None of these

12. The impact of the function $f(z) = iz$ on the complex plane is ______.

(a) translate z by 1 unit

(b) reflects z about the line

(c) rotates z by the right angle counter-clockwise

(d) rotates z by the right angle clockwise

13. The function $f(z) = \frac{1}{\cos z}$ ______.

(a) has not a pole at $z = \frac{\pi}{2}$
(b) is analytic at $z = \frac{\pi}{2}$
(c) is an entire function
(d) None of these

14. If $f(z)$ is analytic within and on a simple closed curve C and z_0 is any point within C, then ______.

(a) $\int_C \frac{f(z)}{(z-z_0)} dz = 0$
(b) $\int_C \frac{f(z)}{(z-z_0)} dz = 2\pi i$
(c) $\int_C \frac{f(z)}{(z-z_0)} dz = 2\pi i f(z_0)$
(d) $\int_C \frac{f(z)}{(z-z_0)} dz = 2\pi i f(0)$

15. The zeros of $f(z) = \cos z$ are ______.

(a) $\frac{(2k+1)\pi}{2}, \quad k \in \mathbb{Z}$
(b) $k\pi, \quad k \in \mathbb{Z}$
(c) $k\pi i, \quad k \in \mathbb{Z}$
(d) $k\pi + i, \quad k \in \mathbb{Z}$

16. The Möbius transformation $\frac{az+b}{cz+d}, a, b, c, d \in \mathbb{C}$ is defined if ______.

(a) $ad - bc < 0$ **(b)** $ad - bc > 0$ **(c)** $ad - bc \neq 0$ **(d)** $ad - bc = 0$

17. Let function $f(z)$ be an analytic function in domain D. Then f is conformal if ______.

(a) $f'(z) = 0$ **(b)** $f'(z) \neq 0$ **(c)** $f'(z) = 0$ **(d)** $f'(z) \neq 0$

18. The Maclaurin series $\sum_{k=0}^{\infty} \frac{z^k}{k!}$ of e^z converges for ______.

(a) for $|z| < 1$
(b) for $|z| \leq 1$
(c) for all $z \in \mathbb{C}$
(d) None of these

19. The period of $f(z) = e^z$ is ______.

(a) π **(b)** $2\pi i$ **(c)** 2π **(d)** π

20. The function $f(z) = z^{2018} + z^{2017} + \cdots + z + 1$ ______.

(a) has at least one zero
(b) has not a non-zero derivative of order 2018
(c) does not satisfy C-R equations
(d) None of these

Candidate's Seat No.________________

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First Semester of M.Sc. (Mathematics) Examination March 2018
MA711 Complex Analysis

Date: 26/03/2018, Monday   Time:10:45 a.m. to 01:00 p.m.   Maximum Marks:50

Instructions:

1. **Section I and Section II must be written in separate answer books.**.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. All notations are standard.

SECTION I

Q - II (A) Attempt any TWO of the following. [**8**]

(1) Show that $u(x, y)$ is harmonic in some domain and find a harmonic conjugate $v(x, y)$ if $u(x, y) = 2x(1 - y)$.

(2) If C_R denote the upperhalf circle $|z| = R$ $(R > 2)$taken in counterclockwise direction. Then show that

$$\left| \int_{C_R} \frac{2z^2 - 1}{z^4 + 5z^2 + 4} dz \right| \leq \frac{\pi R \, (2R^2 + 1)}{(R^2 - 1) \, (R^2 - 4)}.$$

(3) State and prove Morera's theorem.

(4) Find the Maclaurin's series expansion of $7z^3 \, e^{\frac{1}{z}}, \ z \neq 0$.

(B) Attempt any SIX of the following. [**12**]

(1) For $z, w \in \mathbb{C}$, prove that $|z + w| \leq |z| + |w|$.

(2) Find $\text{Arg}(i + 1)$.

(3) Check whether the function $f(z) = |z|$ is analytic or not.

(4) State Liouville's theorem.

(5) Find the fixed points of $w = \dfrac{z - 1}{z + 1}$.

(6) State maximum modulus principle.

(7) State fundamental theorem of algebra.

(8) Evaluate $\int\limits_C e^{3z} \cos z dz$, where C is the contour with vertices $z = 1, z = i, z = 2 + 3i$.

(9) For any $z \in \mathbb{C}$, Prove that $\sin(iz) = i \sinh(z)$.

SECTION II

Q - III (A) Attempt any THREE of the following. [**18**]

(1) State and prove the Cauchy's Theorem.

(2) (i) If C denotes the positively oriented boundary of the square whose vertices are $(\pm 2, \pm 2)$ then evaluate $\int\limits_C \dfrac{e^{-z}dz}{z-(\pi/2i)}$.

(ii) Find the residue of $f(z) = \dfrac{1}{z+z^2}$ at $z = 0$.

(3) Evaluate the integral $\int\limits_{|z|=3} \dfrac{z+1}{z^2-2z} dz$ using residue theorem.

(4) (i) Prove that the Möbius transformation is one-one function.

(ii) Find a bilinear transformation which maps the points $-i, 0, i$ of the z-plane onto the points $-1, i, 1$ of the w-plane respectively.

(5) If a function f is analytic throughout an open disk $|z - z_0| < R_0$ then prove that for each z in this disk, $f(z)$ has a series representation

$f(z) = \sum\limits_{n=0}^{\infty} a_n (z - z_0)^n$ where $a_n = \dfrac{f^{(n)}(z_0)}{n!}$, $\quad n = 0, 1, 2, \ldots$

(B) Attempt any TWO of the following. [**12**]

(1) State and prove Cauchy integral formula.

(2) Show that if $c_1 < 0$, then the image of the half plane $x < c_1$ under the transformation $w = \frac{1}{z}$ is an interior of the circle. What is the image when $c_1 = 0$ under the same transformation?

(3) Expand the function $f(z) = \dfrac{z+4}{(z+3)\ (z-1)^2}$, in Laurent's series expansion valid in (i) $0 < |z-1| < 4$ (ii) $|z-1| > 4$.